

1.1 Similar Projects

Several existing systems have inspired the development of the Skill Barter System. Platforms like LinkedIn Learning, Fiverr, and Skillshare focus on paid services, while platforms such as Simbi promote non-monetary exchange. However, these platforms often lack localized features suitable for Nepalese users, such as language options, community-based reputation systems, and easy barter matching. The Skill Barter System adopts the best features from these global models and tailors them for the local context.

1.1.1 LinkedIn Learning (<https://www.linkedin.com/learning/>)

Thousands of professionally taught online video courses are available on the well-known site LinkedIn Learning. Professional skill development and career promotion are its main priorities. Users can finish courses and use their LinkedIn profiles to highlight their accomplishments. It prevents users from interacting directly or exchanging skills. On the other hand, SkillXchange emphasizes free, peer-to-peer skill sharing, where users teach and learn from one another at no cost, encouraging collaboration and a sense of community.

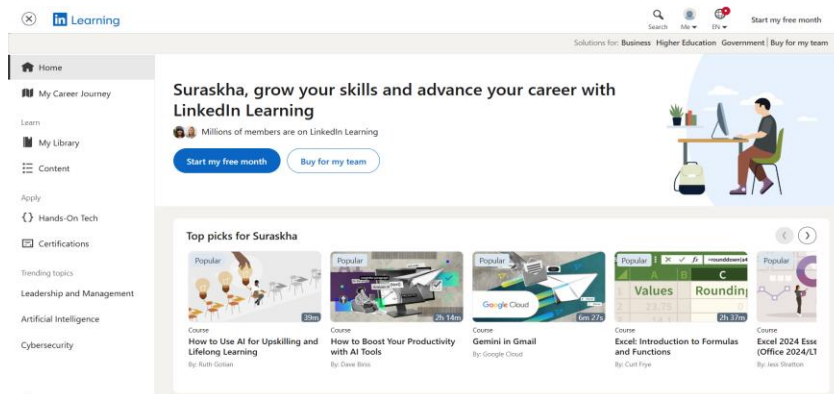


figure 1: LinkedIn Learning's Home Page

Front-end:

- Ember.js (JavaScript framework)
- Core-JS (JavaScript polyfill library)
- Custom UI components (LinkedIn's internal design system)

Back-end:

- Not public

Security:

- HSTS (HTTP Strict Transport Security)
- Cloudflare Bot Management (protection against automation & threats)

CDN:

- Cloudflare

Overview:

LinkedIn Learning uses a highly optimized enterprise-level tech stack designed for speed, security, and heavy user traffic. The presence of Ember.js indicates that the platform uses a single-page application structure, enabling smooth page transitions and fast navigation.

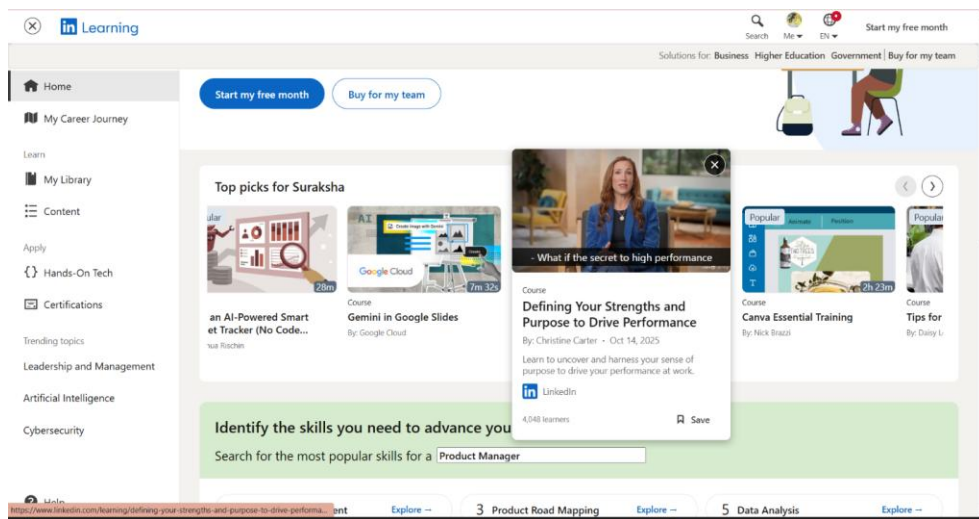


Figure 2: LinkedIn Learning

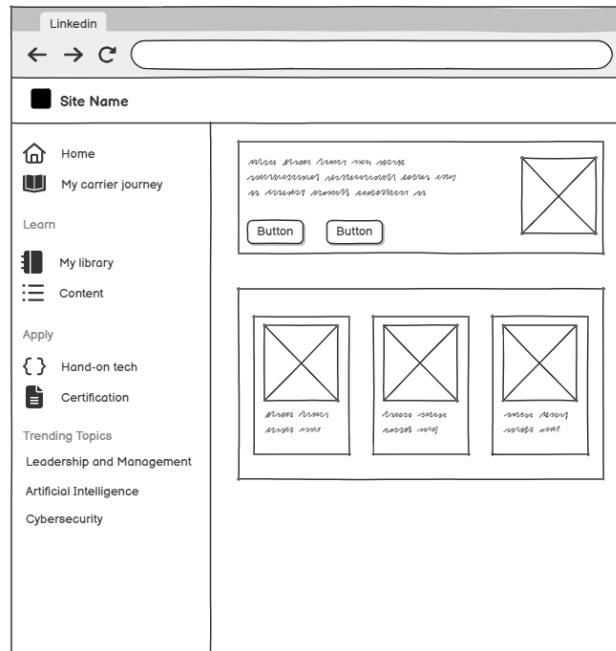


Figure 3: Wireframe of LinkedIn learning

Features I Liked

The platform implements a well-structured card-based listing system. Each course card initially displays a thumbnail and title, while additional metadata becomes visible on hover through a scale transformation. This interaction pattern minimizes visual clutter while ensuring that key information, such as course summary, save action, and learner count, is accessible without leaving the list view.

1.1.2 Skillshare

(<https://www.skillshare.com/en/search/classes?query=freelance>)

SkillShare is another global learning platform that allows experts to create and share video lessons in creative fields like design, photography, and writing. For those who cannot afford to pay for their education, access is limited because all classes require a subscription. SkillShare is focused on one-way learning from teachers to students and lacks teamwork or reciprocal communication. The SkillBarter platform, on the other hand, emphasizes two-way learning by enabling each user to act as both a teacher and a learner. It employs an

AI-based matching engine to connect people with complementary abilities and a gamified system with badges and points to make learning fun and motivating.

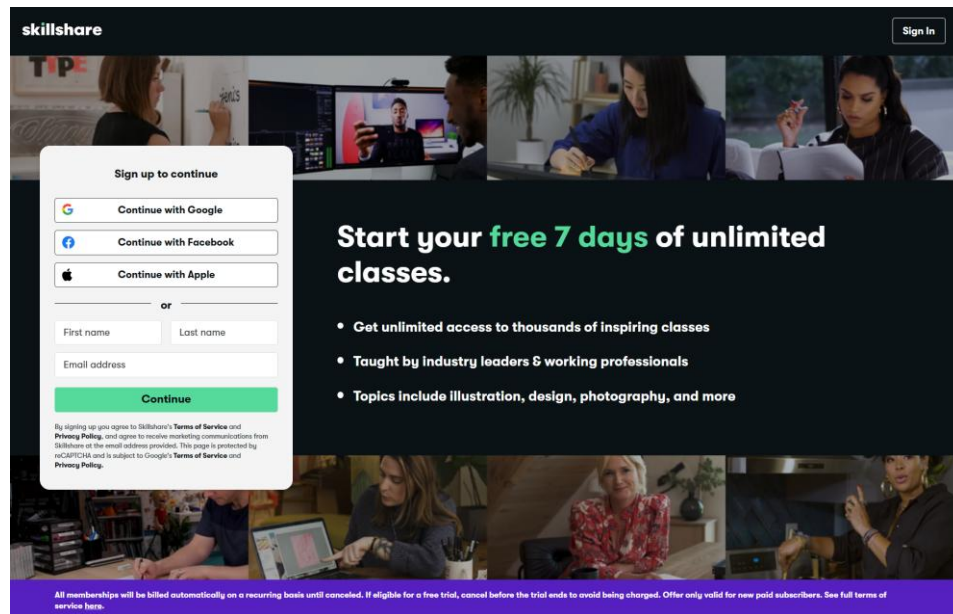


figure 4: Skillshare

Front-end:

- React
- Next.js 12.3.4 (SSR + SSG framework)
- Svelte (used in some modules)
- JSS
- Emotion
- styled components
- Goober (CSS-in-JS library)
- Material UI (UI components)

Back-end:

- PHP (detected through Yii framework)
- Next.js server-side rendering
- AWS (hosting + compute services)

JavaScript Libraries:

- Lodash 4.17.21
- core-js 3.0.1
- jQuery (legacy usage)

Security:

- Cloudflare Bot Management
- HSTS (HTTPS enforcement)

CDN + Server:

- Cloudflare (caching, CDN, security)
- Next.js server environment

Overview:

Skillshare uses a hybrid, modern architecture combining React + Next.js on the front-end with a PHP-based Yii framework on the back-end — showing that the platform is partially migrated to modern infrastructure.

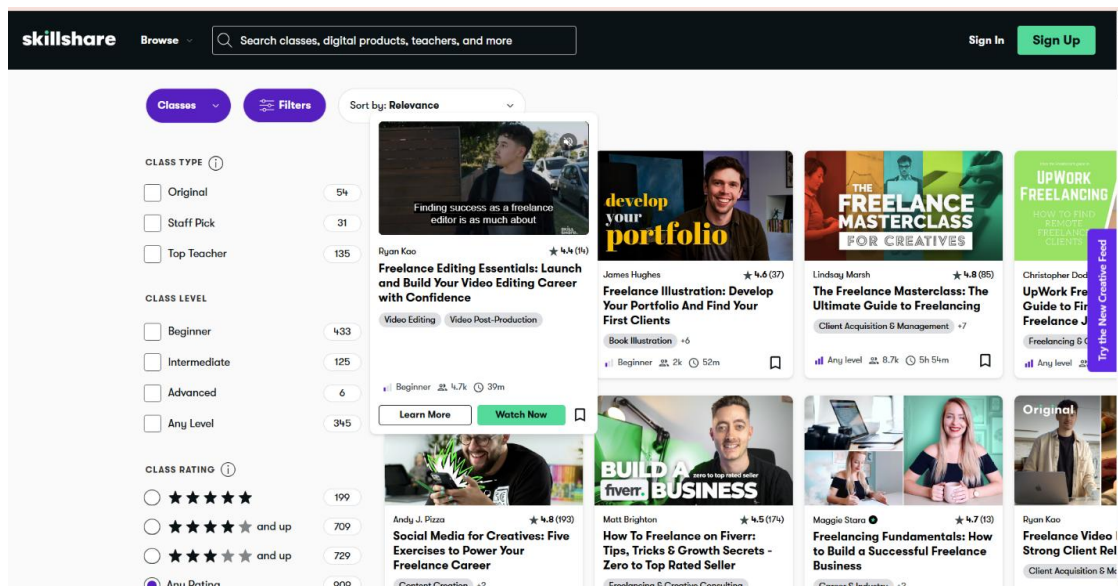


Figure 5: Skill Share

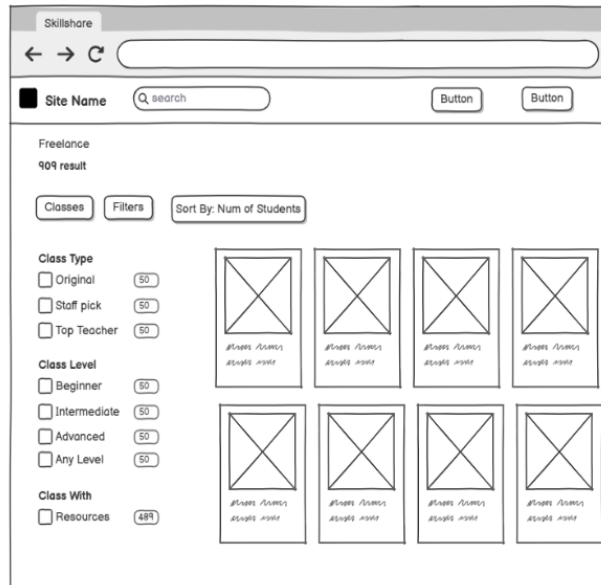


Figure 6: wireframe of Skillshare

Features I liked:

Skillshare provides an efficiently designed browsing layout with filters positioned on the left panel (class type, level, rating) and corresponding course results displayed on the right. This two-column structure enhances discoverability, supports quick refinement of results, and maintains a consistent, visually balanced interface.

1.1.3 Simbi (<https://simbi.com/services/search>)

Simbi is one of the few platforms that allow users to trade services without using money. It works on a credit-based system where users earn credits by offering their skills and spending them to receive services from others. Although Simbi promotes non-monetary exchange, it lacks AI matchmaking, gamification features, and a localized design for users in Nepal. Simbi also does not provide built-in communication or scheduling tools. The SkillXchange system addresses these limitations by introducing an AI skill-matching algorithm, in-app chat and calendar scheduling, and Premium Membership features for users seeking enhanced experience.

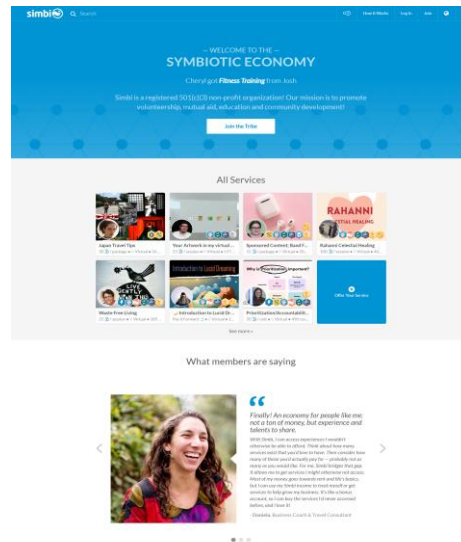


figure 7: Simbi

Front-end:

- Vue.js (primary frontend framework)
- jQuery 4.0.1 (legacy use)
- Lodash 4.17.21
- Hammer.js 2.0.7 (touch gestures)
- core-js 3.32.2 (JS polyfills)
- Bootstrap 3.4.1 (UI layout + components)

Back-end:

- Ruby (primary language)
- Ruby on Rails (main backend framework)
- AWS (hosting infrastructure / PaaS)

Security:

- HSTS (Strict HTTPS Enforcement)

CDN:

- Cloudflare

- cdnjs (JS library delivery)

Overview:

Simbi is built on a classic Ruby on Rails backend paired with a modern Vue.js frontend, providing a balanced architecture that combines reliable backend logic with a responsive, component-based UI.

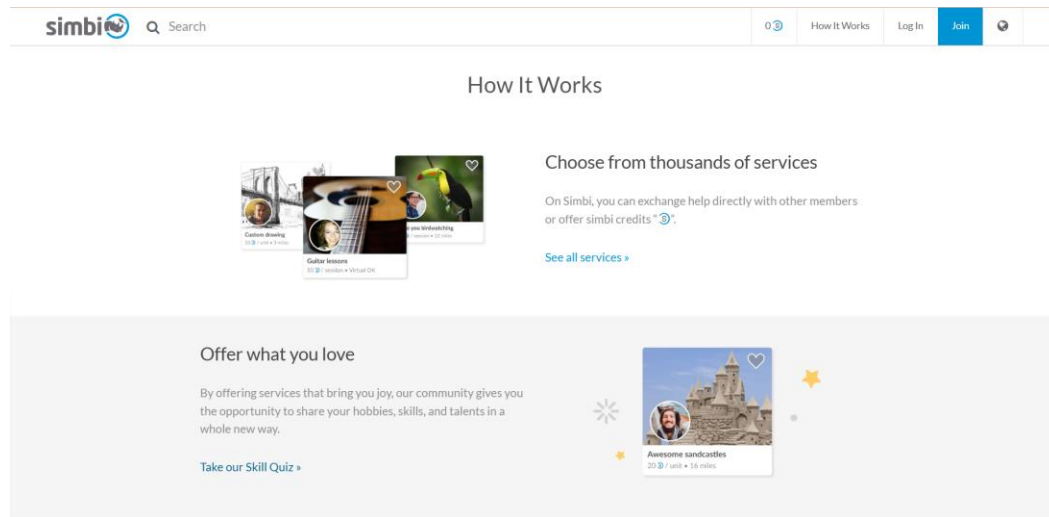


Figure 8: Simbi

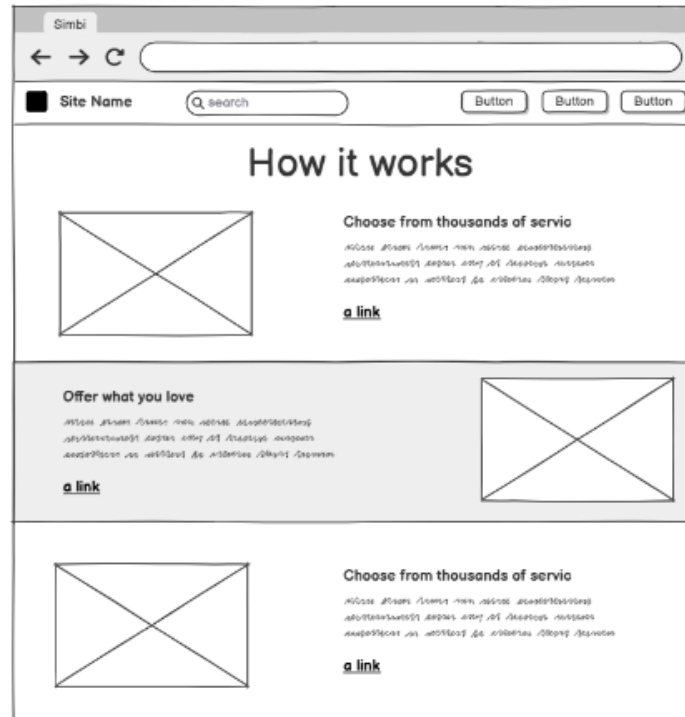


Figure 9: wireframe of Simbi

Featured I liked:

Simbi clearly defines its workflow through a dedicated “How It Works” section supported by contextual links for user guidance. The “Offer What You Love” module is particularly effective as it includes an interactive quiz that helps users identify relevant skills, addressing uncertainty and improving onboarding quality.

1.1.4 PeerList(<https://peerlist.io/scroll>)

Front-end:

- React (primary JavaScript framework for interactive UI)
- Next.js 15.2.3 (modern React-based framework for SSR, SSG, and routing)
- Tailwind CSS (Utility-first CSS framework for fast styling)
- Framer Motion (advanced motion & animation library for UI transitions)
- core-js 3.25.5 (JavaScript polyfills)
- LottieFiles 5.10.2 (animated icons and illustrations)

Back-end:

- Next.js 15.2.3 (also used as backend API routes)
- PHP

Database:

- My SQL

Security:

- HSTS (forces secure HTTPS)
- NextAuth.js (secure authentication framework)
- Google Sign-in (OAuth-based authentication)

Overview:

This platform uses a modern Next.js + React stack combined with Tailwind CSS and Framer Motion for fast, a nimated, and highly responsive UI.

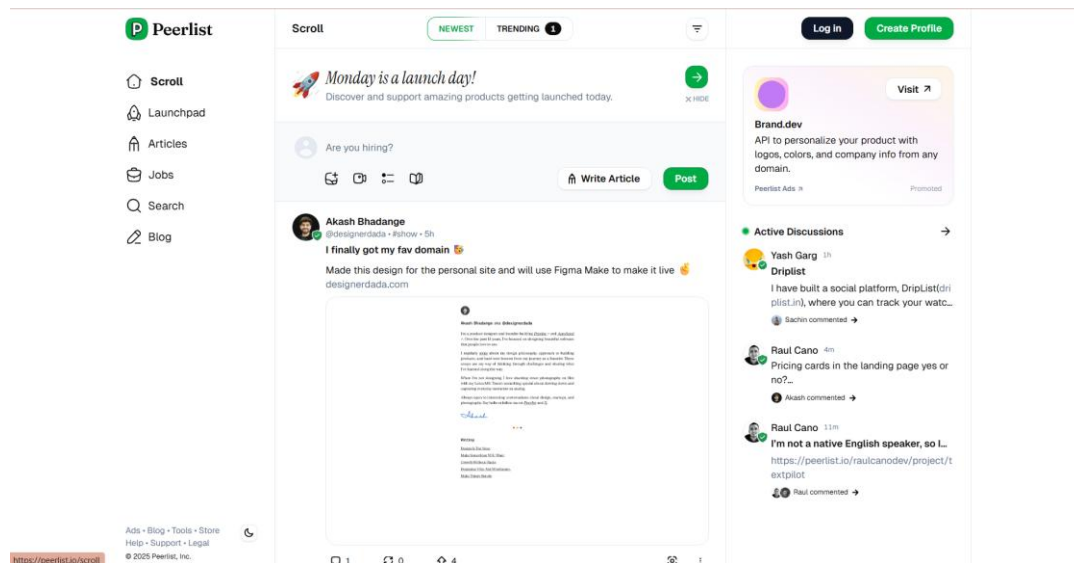


Figure 10:PeerList

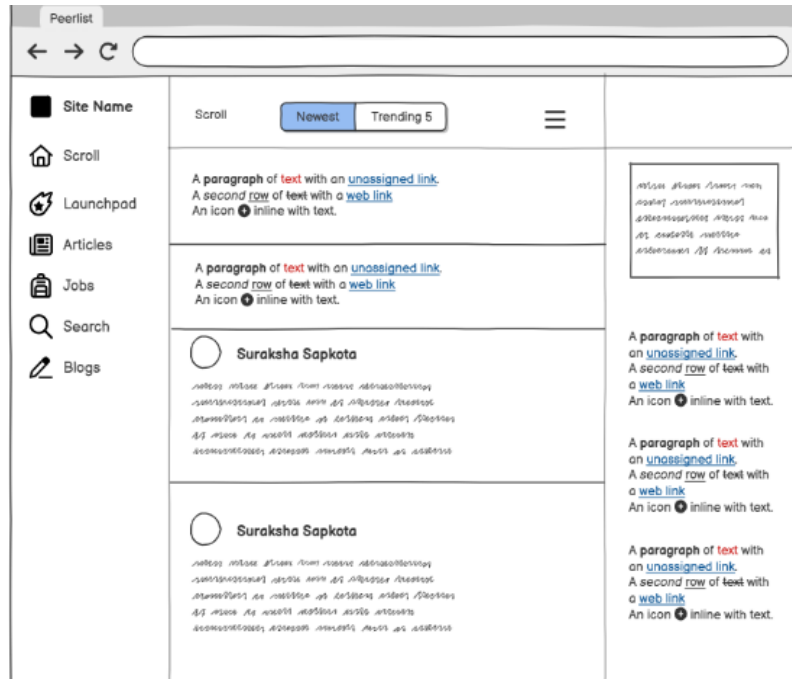


Figure 11: wireframe of peerlist

Featured I liked:

Peerlist has a unique feature where users can share short, useful posts like tips, ideas, and lessons. It also lets people read articles written by other professionals. This makes the platform great for learning and sharing knowledge simply.

1.1.5 Mentorcruise

Front-end:

- React (primary modern JavaScript framework)
- Alpine.js 3.11.1 (lightweight framework used for interactive UI behaviors)
- core-js 3.32.2 (JavaScript polyfills for compatibility)
- jQuery 3.6.0 (legacy script usage for DOM manipulation)

Back-end:

- Django (Python framework) powering dynamic backend logic
- Python as main backend development language
- PHP (used for WordPress-driven sections of the platform)

Security:

- Cloudflare Turnstile (modern captcha alternative for bot protection)
- HSTS (Strict HTTPS for secure communication)
- Cookie Control + Cookiebot 1 (cookie consent + GDPR compliance)
- Cloudflare (DDoS protection, firewall, SSL)

Overview:

This platform uses a hybrid technology architecture, combining:

- Django (Python) for backend application logic
- WordPress (PHP) for CMS content
- React + Alpine.js + HeroUI for responsive UI components
- MySQL for structured data storage

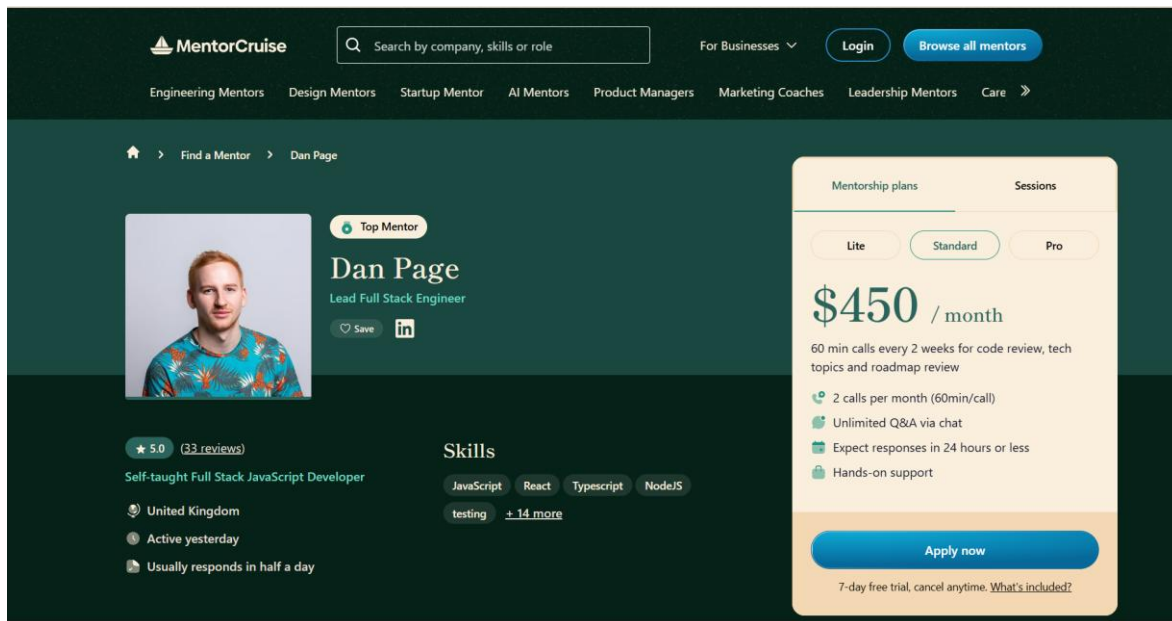


Figure 12: Mentorcruise

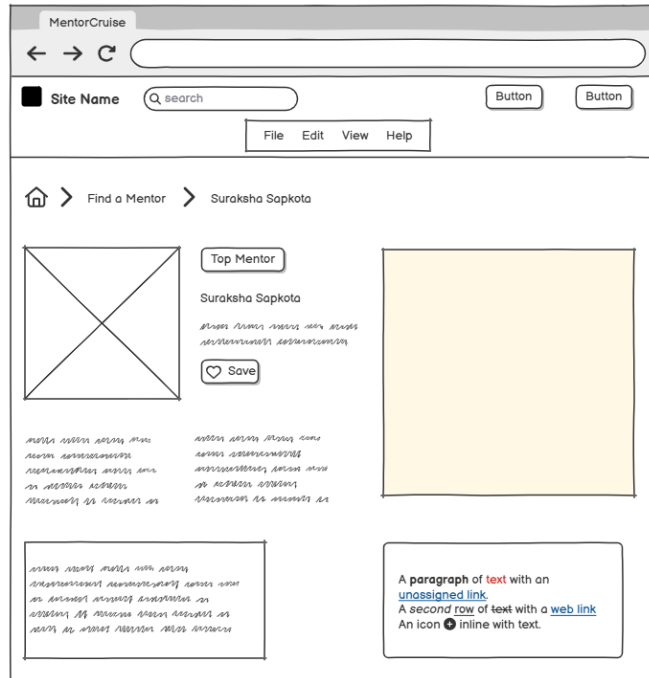


Figure 13: wireframe of mentorcruise

Featured I liked:

The mentor detail page follows a highly optimized information layout, integrating mentor profile images, teachable skills, background description, and user reviews in a single view. This clear hierarchy of information supports efficient decision-making and improves overall discoverability of mentor expertise.